Australian Government
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AUSPOS GPS Processing Report

April 14, 2011

This document is a report of the GPS data processing undertaken by the AUSPOS Online GPS Processing Service (version: AUSPOS 2.01) . The AUSPOS Online GPS Processing Service uses International GNSS Service (IGS) products (final, rapid, ultra-rapid depending on availability) to compute precise coordinates in ITRF anywhere on Earth and GDA94 within Australia. The Service is designed to process only dual frequency GPS phase data.

An overview of the GPS processing strategy is included in this report.

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1 User Data

All antenna heights refer to the vertical distance from the Ground Mark to the Antenna Reference Point (ARP).

Station (s)	Submitted File	Antenna Type	Antenna Height (m)	Start Time	End Time
USER	BR1.11o	TRM41249.00 TZGD	1.056	2011/03/22 19:52:00	2011/03/23 14:02:30

2 Processing Summary

Date	User Stations	Reference Stations	Orbit Type
2011/03/22 19:52:00	USER	AMC2 CHUR DRAO DUBO GODE GOLD MDO1 NRC1 USNO WHIT WILL YELL	IGS final

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3 Computed Coordinates, ITRF2005

All computed coordinates are based on the IGS realisation of the ITRF2005 reference frame. All the given ITRF2005 coordinates refer to a mean epoch of the site observation data. All coordinates refer to the Ground Mark.

3.1 Cartesian, ITRF2005

Station	X (m)	Y (m)	Z (m)	ITRF2005 @
USER	-1911712.758	-4567269.862	4009427.996	22/03/2011
AMC2	-1248596.281	-4819428.205	3976505.976	22/03/2011
CHUR	-236438.976	-3307616.892	5430049.295	22/03/2011
DRAO	-2059164.898	-3621108.403	4814432.302	22/03/2011
DUBO	-417603.799	-4064529.820	4881432.172	22/03/2011
GODE	1130773.656	-4831253.581	3994200.456	22/03/2011
GOLD	-2353614.414	-4641385.286	3676976.413	22/03/2011
MDO1	-1329998.850	-5328393.384	3236504.139	22/03/2011
NRC1	1112777.102	-4341475.874	4522955.861	22/03/2011
USNO	1112189.682	-4842955.038	3985352.298	22/03/2011
WHIT	-2218338.049	-2201205.129	5543057.526	22/03/2011
WILL	-2084258.253	-3313873.004	5019853.032	22/03/2011
YELL	-1224452.793	-2689216.175	5633638.301	22/03/2011

3.2 Geodetic, GRS80 Ellipsoid, ITRF2005

Geoid-ellipsoidal separations, in this section, are computed using a spherical harmonic synthesis of the global EGM2008 geoid. More information on the EGM2008 geoid can be found at http://earth-info.nga.mil/GandG/wgs84/gravitymod/egm2008/

Station	Latitude (DMS)	Longitude (DMS)	Ellipsoidal Height(m)	Derived Above Geoid Height(m)
USER	39 11 18.09838	-112 42 45.44199	1395.079	1415.063
AMC2	38 48 11.24714	-104 31 28.54128	1911.395	1930.141
CHUR	58 45 32.67954	-94 05 19.42716	-19.265	28.900
DRAO	49 19 21.42641	-119 37 29.94007	541.888	558.436
DUBO	50 15 31.71194	-95 51 58.25536	245.297	274.990
GODE	39 01 18.22041	-76 49 36.59227	14.515	47.780
GOLD	35 25 30.56122	-116 53 21.30450	986.653	1017.345
MDO1	30 40 49.83925	-104 00 53.97757	2004.509	2026.588
NRC1	45 27 14.98620	-75 37 25.78526	82.526	116.058
USNO	38 55 08.26700	-77 03 58.41394	48.883	81.750
WHIT	60 45 01.84010	-135 13 19.60770	1427.420	1419.605
WILL	52 14 12.72219	-122 10 04.12755	1095.708	1110.452
YELL	62 28 51.21577	-114 28 50.53270	180.963	207.678

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4 Solution Information

4.1 Coordinate Precision - Geodetic, One Sigma

Station	σ East (m)	σ North (m)	σ Up (m)
USER	0.002	0.002	0.003
AMC2	0.002	0.002	0.003
CHUR	0.002	0.002	0.003
DRAO	0.002	0.002	0.003
DUBO	0.002	0.002	0.003
GODE	0.002	0.002	0.003
GOLD	0.002	0.002	0.003
MDO1	0.002	0.002	0.003
NRC1	0.002	0.002	0.003
USNO	0.002	0.002	0.003
WHIT	0.002	0.002	0.003
WILL	0.002	0.002	0.003
YELL	0.002	0.002	0.003

4.2 Ambiguity Resolution - per baseline

Baseline	Ambiguities Resolved	Baseline Length (km)
DRAO - WILL	92.7 %	370.4
CHUR - YELL	82.0 %	1183.2
CHUR - DUBO	91.5 %	952.2
GOLD - USER	90.2 %	557.9
DRAO - USER	89.7 %	1251.0
WHIT - YELL	87.3 %	1110.9
GODE - USNO	92.3 %	23.7
AMC2 - USER	94.7 %	710.2
WILL - YELL	82.1 %	1227.3
DUBO - NRC1	92.7 %	1596.0
AMC2 - MDO1	87.5 %	901.8
GODE - NRC1	92.5 %	721.0
AVERAGE	89.0%	883.8

Please note for a regional solution, such as used by AUSPOS, an average ambiguity resolution of 50% or better for the network indicates a reliable solution.

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5 Computation Standards

5.1 Computation System

Software	Bernese GPS Software Version 5.0.
GNSS system(s)	GPS only.

5.2 Data Preprocessing and Measurement Modelling

Data preprocessing	Phase preprocessing is undertaken in a baseline by baseline mode using triple-differences. In most cases, cycle slips are fixed by the simultaneous analysis of different linear combinations of L1 and L2. If a cycle slip cannot be fixed reliably, bad data points are removed or new ambiguities are set up A data screening step on the basis of weighted postfit residuals is also performed, and outliers are removed.
Basic observable	Carrier phase with an elevation angle cutoff of 10° and a sampling rate of 3 minutes. However, data cleaning is performed a sampling rate of 30 seconds. Elevation dependent weighting is applied according to $1/\sin(e)^2$ where e is the satellite elevation. The code observable is only used for the receiver clock synchronisation.
Modelled observable	Double differences of the ionosphere-free linear combination.
Ground antenna phase centre calibrations	IGS05 absolute phase-centre variation model is applied.
Tropospheric Model	A priori model is the Saastamoinen-based hydrostatic mapped with the dry-Niell.
Tropospheric Estimation	Zenith delay corrections are estimated relying on the wet-Niell mapping function in intervals of 2 hour. N-S and E-W horizontal delay parameters are solved for every 24 hours.
Tropospheric Mapping Function	Niell
Ionosphere	First-order effect eliminated by forming the ionosphere-free linear combination of L1 and L2.
Tidal displacements	Solid earth tidal displacements are derived from the complete model from the IERS Conventions 2003, but ocean tide loading is not applied.
Atmospheric loading	Not applied
Satellite centre of mass correction	IGS05 phase-centre variation model applied
Satellite phase centre calibration	IGS05 phase-centre variation model applied
Satellite trajectories	Best available IGS products.
Earth Orientation	Best available IGS products.

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5.3 Estimation Process

Adjustment	Weighted least-squares algorithm.
Station coordinates	Coordinate constraints are applied at the Reference sites with standard deviation of 5mm and 10mm for horizontal and vertical components respectively.
Troposphere	Zenith delay parameters and pairs of horizontal delay gradient parameters are estimated for each station in intervals of 2 hour and 24 hours.
Ionospheric correction	An ionospheric map derived from the contributing reference stations is used to aid ambiguity resolution using the QIF strategy
Ambiguity	Ambiguities are resolved in a baseline-by-baseline mode using Quasi-Ionosphere-Free (QIF) approach.

5.4 Reference Frame

Terrestrial reference frame	IGS05 station coordinates and velocities mapped to the mean epoch of observation.
Australian datum	GDA94 coordinates determined via Helmert transformation from ITRF using the Dawson and Woods (2010) parameters.
Derived AHD	For stations within Australia, AUSGeoid09 is used to compute AHD. AUSGeoid09 is the Australia-wide gravimetric quasigeoid model that has been a posteriori fitted to the Australian Height Datum
Above-geoid heights	Earth Gravitational Model EGM2008 released by the National Geospatial-Intelligence Agency (NGA) EGM Development Team is used to compute above-geoid heights. This gravitational model is complete to spherical harmonic degree and order 2159, and contains additional coefficients extending to degree 2190 and order 2159.

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